

Simple Linear Regression

Lecture 14

Topics in this lecture

- Correlation
- Simple linear regression

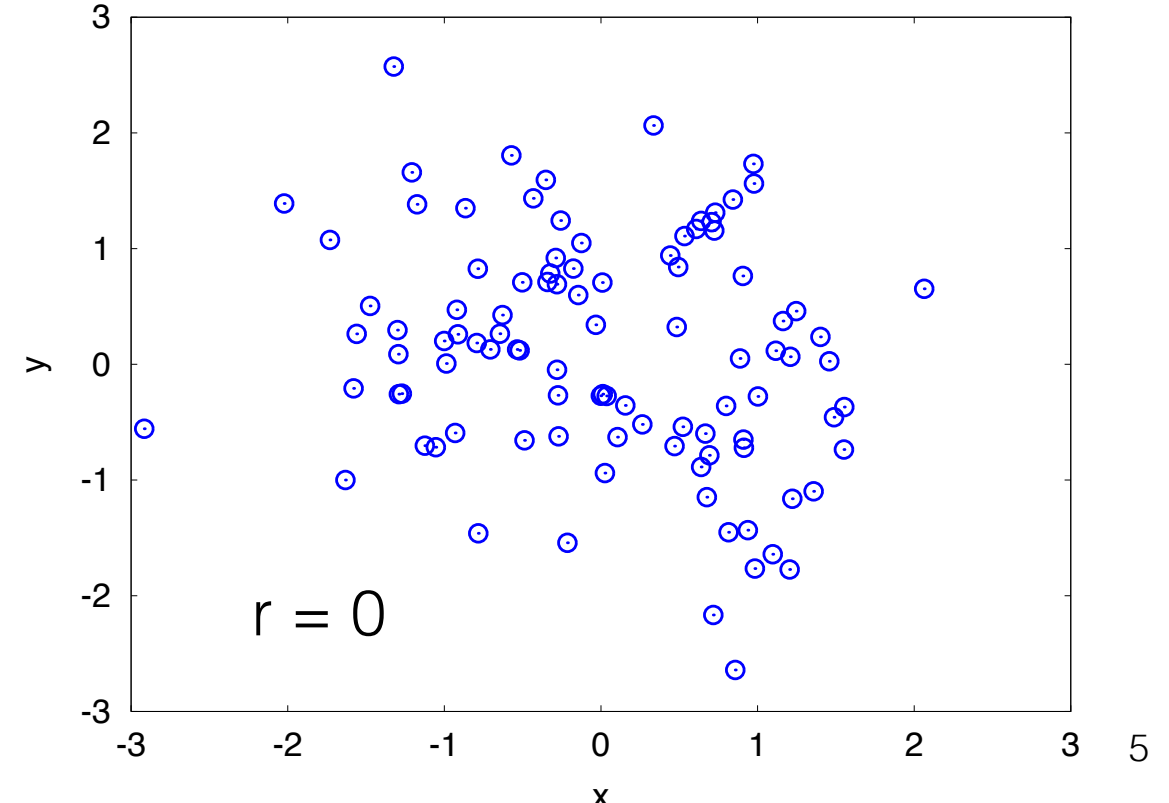
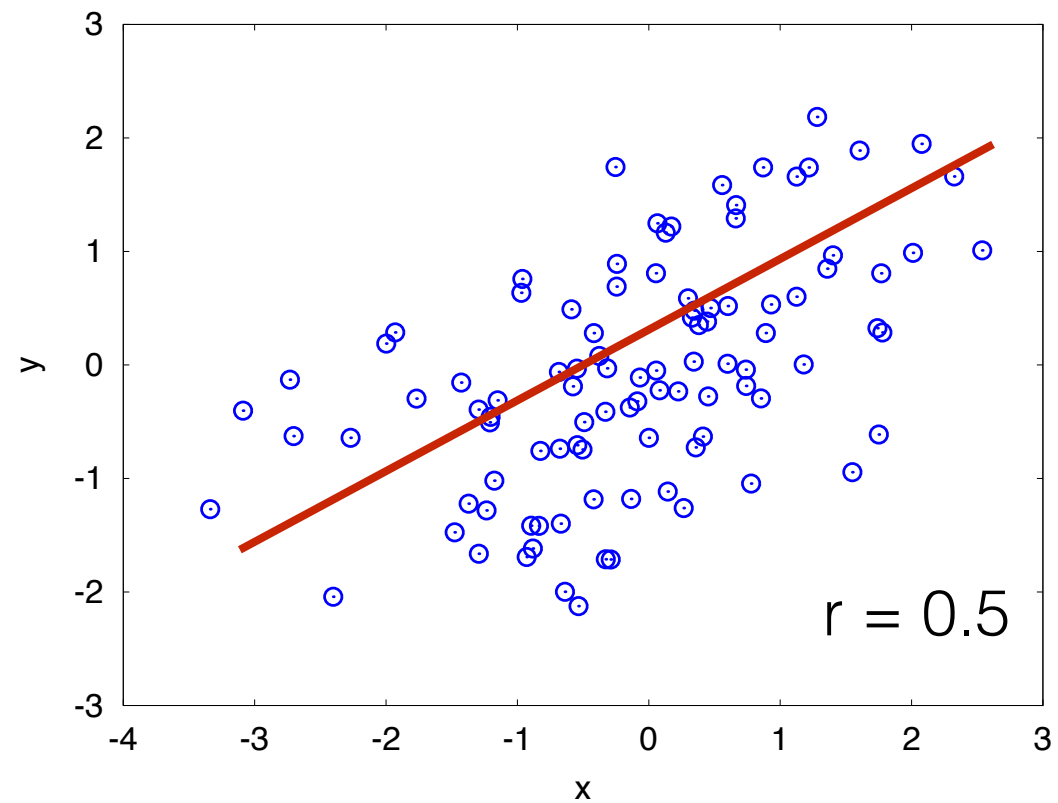
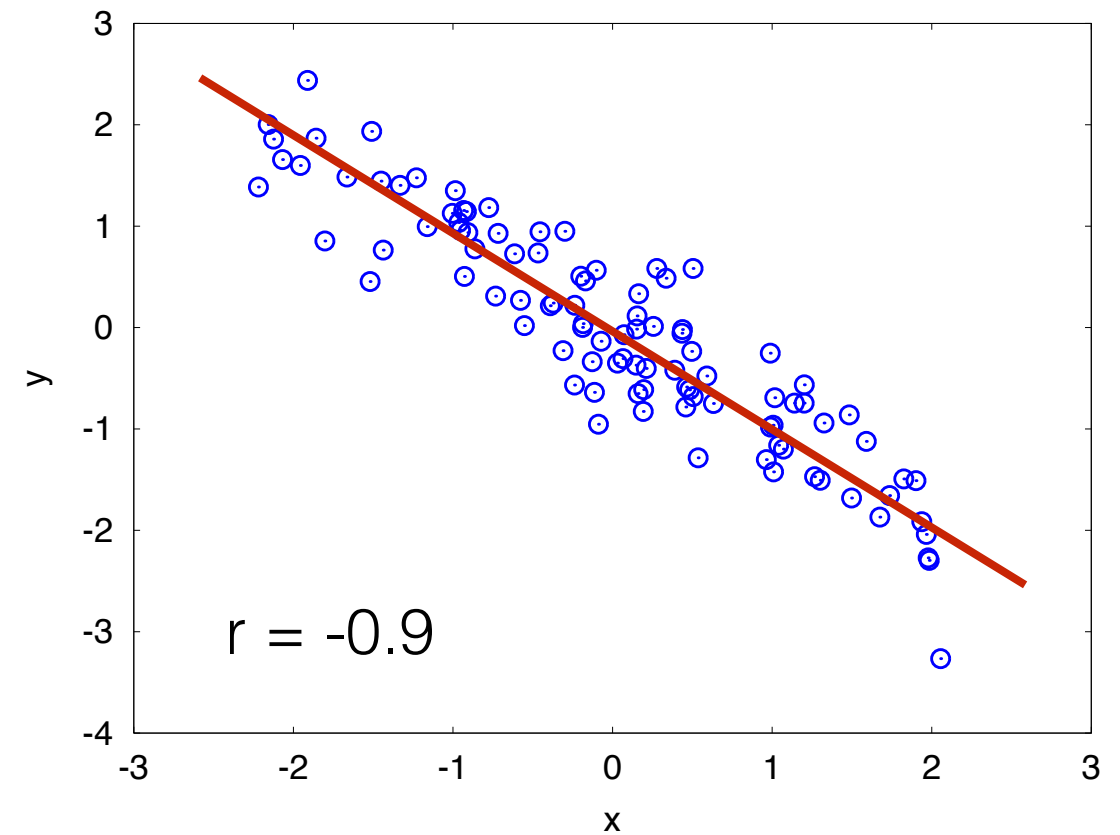
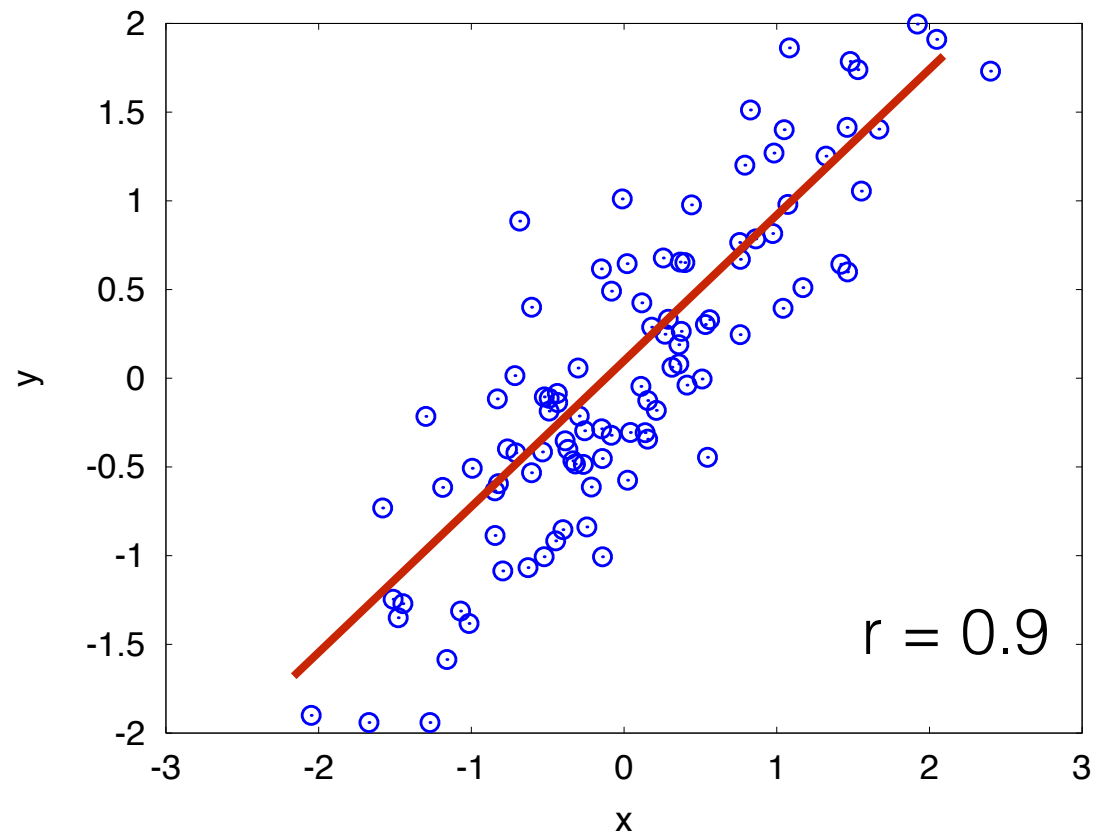
Correlation

- Frequently, we would like to describe a relationship between 2 variables in a data set
- Examples
 - What's the relation between ages of Facebook users and the number of friends they have?
 - What's the relation between height and arm length of a person? Does a taller person tend to have a longer arm length?

Correlation Coefficient

- Correlation coefficient, denoted by r , is a quantity describing how closely related two variables are.
- The value of r indicates the strength of the linear relationship between two variables
- The range of r is from -1 to 1
- If $r = 0$, we say that the two variables are uncorrelated

Values of r



Calculating r

- Given two variables x, y in the data set, the correlation coefficient can be computed as

$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

- Alternatively

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

$$r = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sqrt{\sum_{i=1}^n x_i^2 - n\bar{x}^2} \sqrt{\sum_{i=1}^n y_i^2 - n\bar{y}^2}}$$

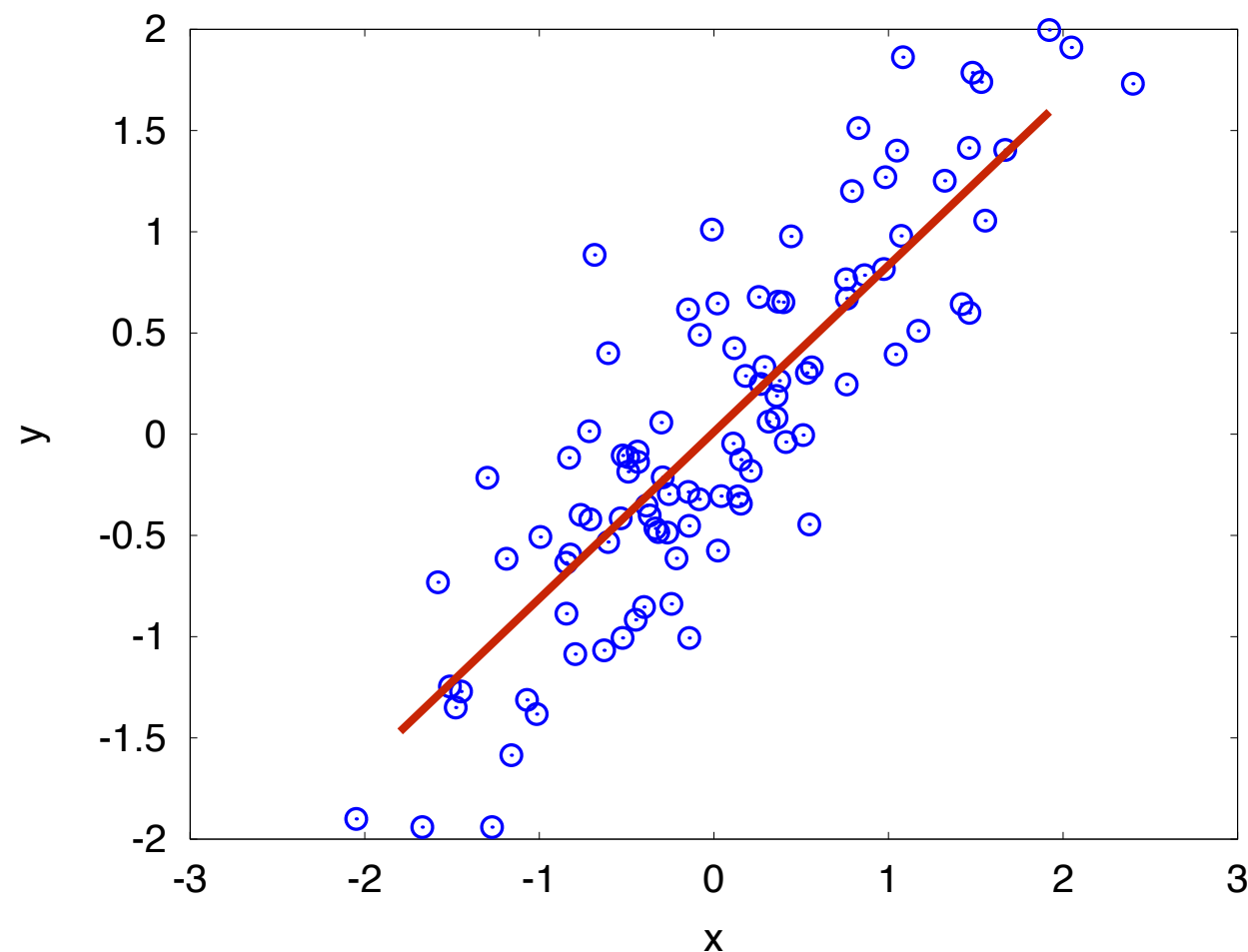
Example 1

- Given two variables x, y in the table. Find the correlation coefficient

x	y
2	4
3	8
7	12
4	9
9	15

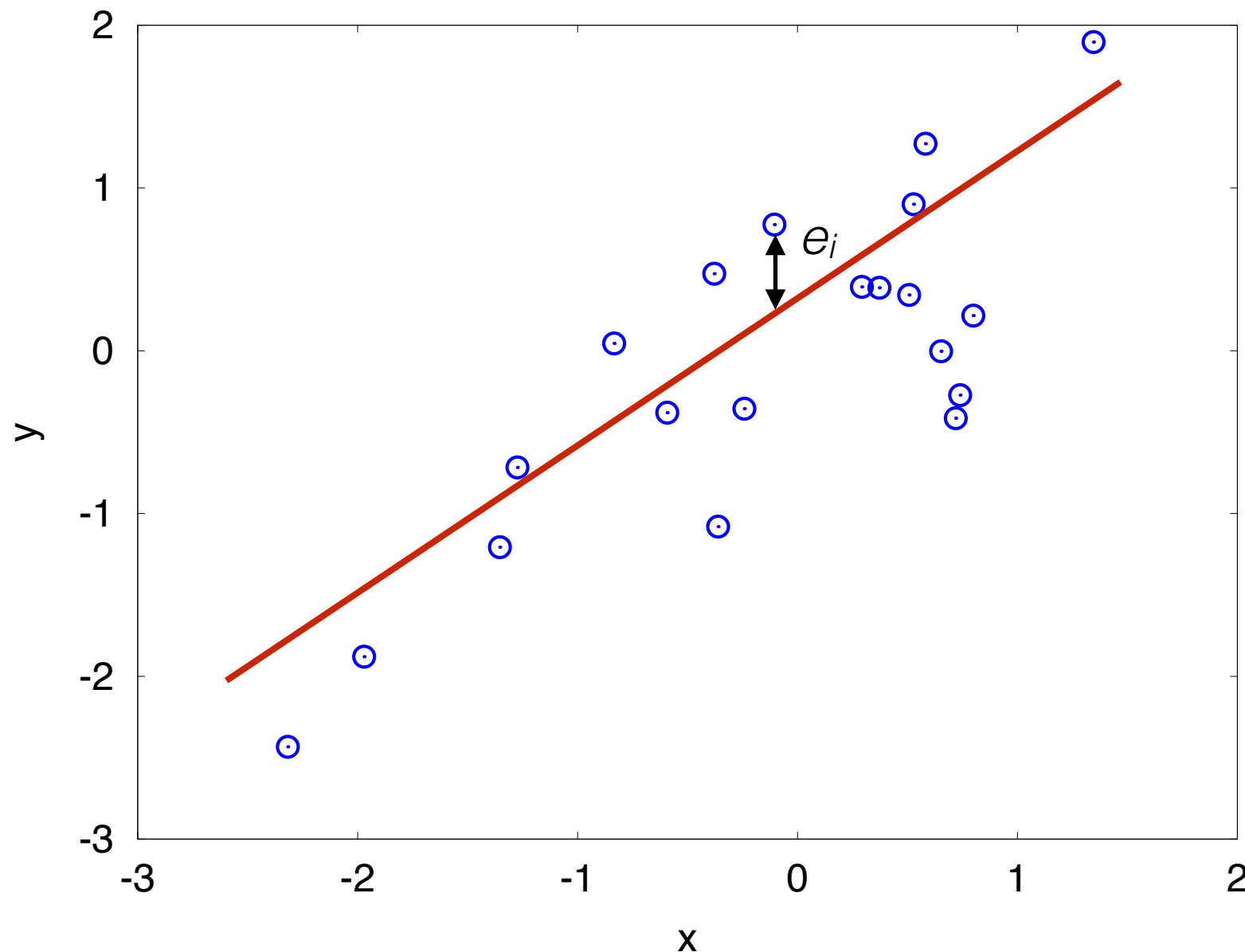
Linear Regression

- Main idea
 - We have 2 variables (x,y)
 - We want to find a line that best fit these data



Least Square Line

- A line that **best fit** the data $\longrightarrow$ minimize $\sum_{i=1}^n e_i^2$



Regression Line

- Actual equation describing a relation between y_i and x_i

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

- Estimated equation

$$y = \hat{\beta}_0 + \hat{\beta}_1 x$$

Best Line

- Residual term

$$e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

- Want to minimize

$$S = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- Solve the following system of equations

$$\begin{aligned}\frac{\partial S}{\partial \hat{\beta}_0} &= - \sum_{i=1}^n 2(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 \\ \frac{\partial S}{\partial \hat{\beta}_1} &= - \sum_{i=1}^n 2x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0\end{aligned}$$

Best Line (2)

- Least square line

$$y = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Example 2

- Given two variables x, y in the table. Find the regression line

x	y
2	4
3	8
7	12
4	9
9	15